

Doom PWAD QA Report

Run d6de8974-9aeb-4cfd-9d0a-652106f12acd | Map E1M1 | Outcome timeout

Executive Summary

This is an autonomous QA test of map E1M1 using a lockstep LLM/MCP test harness, designed to evaluate navigation, combat engagement, resource balance, and secret discovery. The test run ended in a timeout after 47 seconds, failing to complete the map.

Pass/fail:

map_navigation: FAIL

combat_engagement: PASS

resource_balance: PASS

secret_coverage: PASS

overall_verdict: FAIL

- The run ended with outcome 'timeout' after visiting 6 coarse position cluster(s).
- 0 kill(s) recorded against 0 enemy/enemies that spawn at difficulty 1.
- Selected-difficulty health_ratio=0.0000, ammo_ratio=0.0000.
- No secret sectors exist in static analysis.

No technical problems encountered during this test run.

Test Item And Environment

The test evaluated map E1M1 using the Freedoom1 IWAD on difficulty 1. The map contains 8 enemies (Imps, Zombiemen, and a Shotgun Guy), 1 item (Green Armor), 0 keys, and 0 secrets across 5 sectors and 35 linedefs. The map features a relatively simple layout with total dimensions of 2528x1056 units.

The test was conducted using ViZDoom as the game engine, with gemini-3.1-flash-lite as the LLM model via an MCP tool interface. The run was executed at difficulty level 1 with a 600-tick cap.

Metric	Value	Metric	Value
Status	completed	Duration	47
Actions	5	LLM calls	5
Final HP	100	Kills	0
Position samples	305	Movement units	3401.0
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/d6de8974-9aeb-4cfd-9d0a-652106f12acd.mp4		

Static Analysis Context

Metric	Raw WAD	Difficulty 1	Notes
Enemies	8	0	8 hidden by skill/multiplayer flags
Items	1	0	1 hidden by skill/multiplayer flags
Secrets	0	0	Sector special 9
Geometry	35 linedefs	5 sectors	2528 x 1056 units

Selected-difficulty estimate: easy. Health ratio: 0.0. Ammo ratio: 0.0.

Coverage Evaluation

The automated playthrough covered 6 distinct position clusters within the map layout. However, total coverage remained limited as the playthrough timed out without reaching the exit or encountering any enemies. The coverage was insufficient to provide a complete validation of map gameplay.

Objectives Covered

- **Navigate the map (partial exploration):**

Objectives Omitted

- **Navigate the entire map:** Run timed out before completion.
- **Engage all enemy types encountered:** No enemies spawned due to difficulty-specific flags.
- **Find secrets:** 0 secret sectors exist in the map.
- **Stress-test geometry:** Limited exploration prevented thorough interaction with all map features.

Uncovered attributes: The automated playthrough did not encounter any doors, switches, or lifts. The lack of spawned enemies meant that projectile interactions and combat logic were not tested.

Defects And Risks

A total of 1 defect was detected, categorized as Severity 2. The most impactful issue is the difficulty-specific spawn mismatch preventing intended content from appearing during the test.

Patterns: Insufficient defects to identify patterns.

Severity	Priority	Type	Tick	Description
2	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 1, 8 of 8 enemies and 1 of 1 items from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.

Risk Areas

- **Global (Difficulty settings):** medium Map content (enemies/items) is hidden at difficulty 1, rendering the map non-functional for standard difficulty-based QA.

Activity Log Summary

1) Static analysis phase (pre-tick 94), 2) Map exploration phase (ticks 94-536), 3) Report generation phase (post-tick 536).

Exploration was the only activity phase; combat and item collection phases were absent due to spawn issues.

LLM/MCP Decision Trace

#	Ticks	Tool	Stop reason	Reasoning summary
0	14 -> 94	explore	max_tics	Starting position has no immediate threats; initiating exploration to uncover the map layout and potential objectives.
1	95 -> 254	explore	max_tics	The map currently has no visible threats and minimal exploration data; initiating exploration of the south sector to locate objectives or enemies.
2	255 -> 414	explore	max_tics	The map is clear of threats and items in the immediate vicinity. Initiating exploration to progress through the level and find the intended path.
3	415 -> 534	explore	max_tics	The map is clear of enemies and items; initiating exploration to uncover the east sector and find the level exit.
4	535 -> 536	explore	episode_finished	The map remains devoid of enemies and items; initiating exploration towards the east to identify the exit or further progression path.

Limitations And Recommendations

The automated playthrough did not reach any locked doors or specialized geometry triggers.

Good Quality Areas

- **Exploration:** The automated playthrough demonstrated consistent movement across the layout without getting stuck in geometry.